

Project layout, archives, sed and diff

A design project needs a clear tree, source, testbench, docs, scripts, and a build folder for generated noise

Tree, search, edit, compare

- Keep RTL under src, tests under tb, notes under docs, helpers under scripts
- Find locates files by name; recursive grep finds a signal across the tree
- Sed substitutes on a stream, test without in-place first
- Unified diff is the same shape as git diff and feeds patch
- When you archive, ignore build and logs so the tarball stays small and shareable

Browser lab

Digital Design and Verification Platform

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Inspect a design-project tree, search with find/grep patterns, and preview what tar/zip would pack (lab simulation).

Starter example: find *.v, then tar -czf with ignore rules (skip build/ & logs/).

Load starter example

Challenges 0 / 22 cleared

Find *.vc find all Verilog files (*.v).

Show hintCheckNextIdle

Find *.vFind *.mdFind scriptsFind logs/*Find buildgrep modulegrep ERRORgrep sample_projectgrep bashtar respect ignoretar everythingPacked count (ignore)sed module-MODULEsed on aluLab-LABdiff main vs alupatch previewIgnore *.oIgnore logFind .gitignoregrep INFOFind README

Project tree

sample_project/
[file] README.md
[dir] src/
[file] main.v
[file] alu.v
[dir] tb/
[file] test_main.v
[dir] scripts/
[file] run_demo.sh
[dir] docs/
[file] notes.md
[dir] build/
[file] main.o (ignored)

find / grep

*.vfind

modulegrep -R

src/main.v
src/alu.v
tb/test_main.v

Archive preview

tar czf (respect ignore)tar czf (everything)

sample_project.tar.gz (7 files)

keep README.md (420 B)
keep src/main.v (80 B)
keep src/alu.v (120 B)
keep tb/test_main.v (200 B)
keep scripts/run_demo.sh (160 B)
keep docs/notes.md (90 B)
keep .gitignore (40 B)

skipped (ignore rules):

sed & diff / patch

s/module/MODULE/

src/main.v

sed

diff main.v vs alu.v

apply patch (preview)

Project archives lab

Real shell practice

```
wsl - module19-project-archives/examples/project_structure

# Track A - real Unix
# real shell session (Track A)

$ ls sample_project
build
docs
scripts
src
tb

$ find sample_project -type f
sample_project/build/.gitkeep
sample_project/docs/notes.md
sample_project/scripts/build.sh
sample_project/src/main.v
sample_project/tb/test_main.v

$ find ../find_grep -name '*.v'
../find_grep/src/top.v
../find_grep/src/util.v
../find_grep/tb/tb_top.v

$ grep -Rn CLOCK_SIGNAL ../find_grep --include='*.v'
../find_grep/src/top.v:2:// Contains CLOCK_SIGNAL for grep -R practice.
../find_grep/src/top.v:4:  input  wire CLOCK_SIGNAL,
../find_grep/src/top.v:8:  always @(posedge CLOCK_SIGNAL or posedge rst) begin
../find_grep/src/util.v:2:// Does not contain CLOCK_SIGNAL.
../find_grep/tb/tb_top.v:2:// Contains CLOCK_SIGNAL for grep -R practice.
../find_grep/tb/tb_top.v:4:  reg  CLOCK_SIGNAL;
../find_grep/tb/tb_top.v:7:  top u_dut (.CLOCK_SIGNAL(CLOCK_SIGNAL), .rst(rst), .out(out));
../find_grep/tb/tb_top.v:9:    CLOCK_SIGNAL = 0;
../find_grep/tb/tb_top.v:10:    forever #5 CLOCK_SIGNAL = ~CLOCK_SIGNAL;

$ sed 's/old/new/g' ../sed_basics/sample.txt
line one new text
line two new and new again
line three
new at start

$ diff -u ../diff_patch/original.txt ../diff_patch/modified.txt
--- ../diff_patch/original.txt  2026-07-18 16:34:55.486570800 +0800
+++ ../diff_patch/modified.txt  2026-07-18 16:34:55.482542200 +0800
@@ -1,4 +1,5 @@
 Line one
-Line two
+Line two (edited)
 Line three
 Line four
+Line five added
```

Real shell — layout find sed diff

Real shell practice — try these

```
# ls / find — inspect sample_project layout
ls sample_project
```

```
find sample_project -type f
```

```
# find *.v — Verilog files under find_grep
find ../find_grep -name '*.v'
```

```
# grep -Rn — recursive content search with line numbers (Verilog only here)
grep -Rn CLOCK_SIGNAL ../find_grep --include='*.v'
```

```
# sed 's/old/new/g' — global substitute (stdout only; file unchanged)
sed 's/old/new/g' ../sed_basics/sample.txt
```

```
# diff -u — unified diff (patch-friendly)
```

Pitfalls to watch

- Do not archive build and logs by default, they bloat shares and hide the real sources
- Always dry-run sed without in-place before you rewrite a file
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, walk the sample project, practice find and grep, then sed and diff
- When you are ready, take the short quiz, then continue to tar versus zip